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28 OCTOBER 1960
SUPERSEDING
MIL–E–11450
13 SEPTEMBER 1951

MILITARY SPECIFICATION

EVACUATION BAG, CASUALTY, INSULATED, WITH LINER AND CARRYING BAG

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers requirements for a casualty evacuation bag of one type and size. It is of quilted construction and insulated with waterfowl feathers and down. In addition, two insulating pads of synthetic pile which provide added insulation under a casualty at the principal pressure points are integral components of the bottom of the bag. The bag is provided with a removable liner which has a moisture proof bottom and a vapor permeable top. A waterproof carrying bag is included for transporting the rolled bag and to protect it when not in use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

C–F–160 — Feathers, Waterfowl, and Down, Waterfowl.
L–P–00378 — Polyethylene Film, Thin Gauge.
V–F–106 — Fasteners, Slide, Interlocking.
V–T–276 — Thread, Cotton.
CCC–T–191 — Textile Test Methods.
DDD–L–0020 — Labels, Printed and Markings, General Use (For Clothing, Equipage and Tentage).
DDD–T–86 — Tape, Cotton, Bleached, Dyed or Grey, General Use.
PPP–B–636 — Box, Fiberboard.
PPP–T–60 — Tape, Pressure-Sensitive, Adhesive, Waterproof, for Packaging and Sealing.

MILITARY

MIL–C–332 — Cloth, Cotton, Balloon and Percale.
MIL–B–371 — Braid, Textile (Tubular).
MIL–W–530 — Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL–B–543 — Buckles, Tongueless and Web Strap.
MIL–B–3108 — Bag, Waterproof, Clothing.

FSC 6530

571047—61—(662)

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MIL-T-3530 — Treatment, Mildew-Resistant and Water-Repellent for Thread and Twine.
MIL-C-3924 — Cloth, Cotton Warp and Nylon Filling, Oxford.
MIL-C-5040 — Cord, Nylon.
MIL-C-7219 — Cloth, Nylon, Duck, Plied Yarns, for Parachute Packs.
MIL-T-7807 — Thread, Nylon.
MIL-C-10772 — Cloth, Rayon, Acetate (Saponified), Rip Stop.
MIL-F-10884 — Fasteners, Snap.
MIL-F-11861 — Fur Strips (Wolverine and Wolf).
MIL-C-14366 — Cloth, Coated, Nylon, Polyvinyl Butyral.
MIL-G-16491 — Grommets, Metallic.
MIL-E-20652 — Eyelets and Washers.
MIL-C-40049 — Cloth, Nylon, Taffeta, 2.5 Ounce.

STANDARDS

FEDERAL

Fed. Std. No. 751 — Stitches, Seams, and Stitchings.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 — Marking for Shipment and Storage.

DRAWINGS

QUARTERMASTER CORPS

2-3-123 — Evacuation Bag, Casualty, Insulated, with Liner and Carrying Bag; Assembly, Views and Sections.
2-3-124 — Evacuation Bag, Casualty, Insulated, with Liner and Carrying Bag; Bag Details.
2-3-125 — Evacuation Bag, Casualty, Insulated, with Liner and Carrying Bag; Strap Details.
2-3-126 — Evacuation Bag, Casualty, Insulated, with Liner and Carrying Bag; Sections.
2-3-127 — Evacuation Bag, Casualty, Insulated, with Liner and Carrying Bag; Liner Assembly and Details.
2-3-128 — Evacuation Bag, Casualty, Insulated, with Liner and Carrying Bag; Assembly, Details and Sections.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Figures 1 through 5 are miniature reproductions of referenced drawings, and are attached for information only.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

COLOR ASSOCIATION OF THE UNITED STATES, INC.

United States Army Color Card of Standard Shades for Slide Fastener Tapes.

(Application for copies should be addressed to the Color Association of the United States, Inc., 200 Madison Ave., New York 16, N. Y.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Cloth, cotton warp and nylon filling, oxford.* The outer casing and insulating pad binding strips shall be cloth, cotton warp and nylon filling, oxford, 5 ounce, olive green No. 107, water repellent, conforming to Specification MIL-C-3924.

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3.1.2 *Cloth, cotton, balloon.* The inner casing shall be cloth, cotton, balloon, 2.8 or 3.9 ounce, shade olive drab No. 3, water repellent, preshrunk, shall have satisfactory down retention qualities and conform to type I or II, class 2 of Specification MIL–C–332.

3.1.3 *Cloth, rayon, acetate (saponified), rip stop.* The diaphragms shall be cloth, rayon, acetate (saponified), rip stop, olive green No. 106, conforming to Specification MIL–C–10772.

3.1.4 *Cloth, nylon, duck.* The drag pad on the bottom of the bag shall be cloth, nylon, duck, 9.5 ounces per square yard, conforming to type I of Specification MIL–C–7219, and dyed olive green No. 106 to match the standard shade sample (see 6.3 and 6.4).

3.1.5 *Cloth, nylon, taffeta.* The top of the liner shall be cloth, nylon, taffeta, 2.5 ounce, olive green No. 106, conforming to Specification MIL–C–40049.

3.1.6 *Cloth, coated, nylon, polyvinyl butyral.* The bottom of the liner shall be cloth, coated nylon, polyvinyl butyral, 3.5 ounces per square yard, conforming to Specification MIL–C–14366, except that the color of the face side of the fabric may approximate, but need not match, shade taupe 179.

3.1.7 *Webbing, textile, cotton, type IIa, 1 inch and 1½ inch.* The webbing shall be shade olive drab No. 7, type IIa, class 4, conforming to Specification MIL–W–530, except that it shall not be mildew-resistant treated.

3.1.8 *Tape, textile, cotton, ¾ inch and 2½ inch.* The tape shall be shade olive drab No. 7, type I, class 4, conforming to Specification MIL–W–530, except that it shall not be mildew-resistant treated.

3.1.9 *Tape, cotton, woven, 1 inch.* The liner loops on inside of bag, tie tapes and snap fastener reinforcement on the liner, shall be tape, cotton, woven, shade olive drab No. 7, conforming to class 3 of Specification DDD–T–86.

3.1.10 *Fur strips.* The fur strips used in fabrication of the fur ruff shall conform to Specification MIL–F–11861. Preference in procurement shall be given to available types in the following order: Type IA, type II, type IIIA, and type III. Strip and hair length shall be as specified for class 2.

3.1.11 *Feathers and down.* The insulating filling material shall be a mixture of waterfowl feathers and down, or pure down as indicated on drawings. The mixture shall conform to type III and the down to type II, class 1 or 2, grade A or Specification C–F–160.

3.1.12 *Thread, cotton.* The thread shall conform to Specification V–T–276. Sizes and types shall be as follows:

For all stitching except that used in the liner, forming longitudinal channels, and where optional stitch type 401 is used.	— Type IA3, ticket No. 30, 3 ply for needle and bobbin.
For stitching longitudinal channels except with optional stitch type 401.	— Type IA3, ticket No. 50, 3 ply for needle and bobbin.
For stitching longitudinal channels with optional stitch type 401.	— Type IA3, ticket No. 50, 3 ply for needle; ticket No. 70, 2 ply for looper.
For optional stitch type 401 in other than longitudinal channels.	— Type IA3, ticket No. 30, 3 ply for needle; ticket No. 50, 3 ply for looper.

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3.1.12.1 *Color and treatment.* The thread shall be shade S–1 (Cable No. 66022) and shall be water-repellent treated in accordance with type II of Specification MIL–T–3530. The dyed thread shall show fastness to laundering equal to or better than the standard sample (see 6.3).

3.1.13 *Thread, nylon.* The thread for stitching the removable liner shall be shade S–1 (Cable No. 66022) and conform to type I, class 2, size E of Specification MIL–T–7807.

3.1.14 *Cord, nylon.* The drawcord of the carrying bag shall conform to type III, table II of Specification MIL–C–5040.

3.1.14.1 *Color.* The color shall be olive green No. 106 matching a standard shade sample (see 6.3). Both ends of the drawcord shall be seared.

3.1.15 *Grommets.* The grommets used in the carrying bag shall be type III, class 3, size 0, conforming to Specification MIL–G–16491.

3.1.16 *Dynel pile fabric.* The dynel pile fabric used for making the insulating pads shall be circular knitted with a cotton backing yarn in each feed. The backing fabric shall have not less than 18 courses and 12 wales per inch. The dynel pile portion of the fabric shall be evenly sheared, and shall consist of a blend of approximately equal parts of 12 and 24 denier cut dynel staple. The staple length shall be such as to produce a finished fabric 1¼ ± ⅛ inches thick when tested using a thickness gage with a foot having a minimum area of 16 square inches and exerting a load of 0.005 psi. The finished pile fabric shall weigh not less than 32 nor more than 35 ounces per square yard and shall analyze not less than 85 percent dynel when tested as specified in 4.2.1.

3.1.17 *Fastener, slide, interlocking.* The slide fastener shall conform to size MH, class A of Specification V–F–106. Slide fastener chains shall be fabricated of natural finish aluminum, natural finish brass or zinc alloy finished in accordance with manufacturer's standard practice. All other metal components shall be fabricated of brass, nickel silver or zinc alloy except slider bodies which shall be fabricated of die cast zinc only. All metal components shall approximately match each other in color except as stated below.

3.1.17.1 *Finish.* Nickel plate, cadmium plate and zinc plate are considered satisfactory color matches for aluminum chain. Zinc alloy components and slider pulls, when furnished with aluminum chain, may be furnished with an olive drab chromate conversion coating in lieu of the color match required above. Bronze chromate finish on zinc alloy components and slider pulls is considered a color match for natural finish brass.

3.1.17.2 *Arrangement of stops, chain and sliders.* The arrangement and location of stops, chain and sliders shal be as shown on the drawing. The ends of the fastener shall be equipped with open end stops. The stops shall be located to allow as complete a closure as practicable without interference between the two end sliders. The length of the slide fastener chain shall be such that after the slide fastener is sewed on, the tape shall not be puckered and the sliders shall operate freely.

3.1.17.3 *Sliders and pulls.* The slide fastener shall be equipped with six sliders to provide an opening at any point along the chain. The two head section sliders shall be non-locking and of the reversible type, equipped with single wire stirrup pulls and thongs on each side. The remaining four sliders shall be nonlocking and of the non-reversible type, equipped with single wire stirrup pulls and thongs operable from the outside of the bag.

3.1.17.4 *Slide fastener tape.* The slide fastener tape shall be water repellent treated, shade S (Cable No. 66519) of the U. S. Army Color Card of Standard Shades for Slide Fastener Tapes (see 2.2) and shall show good fastness to laundering when tested as specified in 4.2.1.

3.1.17.5 *Braid thong.* Each stirrup pull shall be equipped with a flat tubular braid thong, looped through the opening of the slider pull and knotted at one end. The thong shall finish 3½ (+ 1, —½) inches long. The braid for the thong shall be cotton, tubular, flat, ⅜ inch, water-repellent, shade olive green No. 107, conforming to type VII of Specification MIL–B–371.

3.1.18 *Fastener, snap, brass, black chemical finish, male and female.* The snap fasteners shall

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be style 2A, finish 2, construction A, conforming to Specification MIL–F–10884.

3.1.19 *Buckles, tongueless, 2 bar.* The buckles shall be ¾ inch, type II, class 1, construction B, conforming to Specification MIL–B–543, and have a black chemical finish.

3.1.20 *Eyelets.* The eyelets shall be No. 4132 conforming to Specification MIL–E–20652.

3.1.21 *Wire.* The wire stiffener for the fur ruff shall be bright copper, 0.080 ± .003-inch diameter, dead soft, so that when bent 360° over a mandrel of the same diameter and the wire straightened, and the bending and straightening process repeated at the same place on the wire, the wire shall not crack or break.

3.1.22 *Bag, carrying.* The carrying bag shall conform to class 1, of Specification MIL–B–3108, except that the design of the bag and contractor's label shall be in accordance with Drawing 2–3–128 and as specified herein.

3.1.23 *Marking.* All marking shall conform to Specification DDD–L–0020.

3.1.23.1 *Identification marking.* The identification marks on the evacuation bag and liner shall conform to type IV, class 5, and shall be located as shown on the drawings.

3.1.23.2 *Laundry instructions.* Laundry instructions for the evacuation bag and liner shall conform to type IV, class 9, and shall be located as shown on the drawings. Instructions shall be printed in characters not less than $\frac{3}{16}$ inch in height with appropriate increase in size of captions. Instructions shall be as follows:

"To be laundered in accordance with formula "G", D/A Technical Manual TM 10–354

Do Not Dry Clean"

3.1.23.3 *Label for carrying bag.* A contractor's label approximately 4 inches by 4 inches, conforming to type II, class 9, shall be stitched to the outside of the bag, along the lower edge of the drawcord channel in the location shown on the drawing. The printing of the first three lines shall be in characters not less than $\frac{3}{16}$ inch high and shall be as follows:

"Carrying Bag for Evacuation Bag, Insulated, with Liner

Stock number (see 6.2)

Contract number and date"

In addition, the following shall be printed in characters not less than $\frac{3}{32}$ inch high:

"The Bag and Liner contained herein are to be used only for evacuation or protection of casualties. Do not use as sleeping bag."

3.1.23.4 *Other marking.* The letters U. S. and the words "For casualties only" on the evacuation bag and liner, shall conform to type IV, class 9 in the size characters and locations shown on drawings.

3.1.23.5 *Content label.* A content label, conforming to type I, class 9, meeting the state requirements for the insulating fillers shall be placed at the foot end of the evacuation bag in the location shown on the drawing.

3.1.23.6 *Fastness.* All printing and marking shall show fastness to laundering as specified for class 6 labels.

3.1.24 *Patterns.* The patterns shall be made by the contractor in accordance with dimensions shown on drawings.

3.2 Construction. The construction shall conform in all respects to the drawings listed in section 2 and as specified herein.

3.2.1 *Stitching, machine.* Stitching, except bartacking, shall conform to Federal Standard No. 751 as follows:

For all stitching, except as specified below.	— Type 301, 10 to 14 stitches per inch.
Optional method in lieu of 301, as indicated by Z on drawings.	— Type 401, 10 to 14 stitches per inch.
Sewing and piecing of webbing, seaming insulation pads to bottom of bag, stitching fur ruff, seaming liner and carrying bag.	— Type 301, 8 to 10 stitches per inch.

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3.2.1.1 *Type 301 stitching.* Thread breaks in stitching shall be backstitched not less than 1 inch at each break. Stitching ends shall be backstitched not less than ½ inch, except where ends are turned under in a hem or held down by other stitching. Thread tension shall be maintained so that there will be no loose stitching and the lock will be embedded in the center of the materials sewed.

3.2.1.2 *Type 401 stitching.* Thread breaks in stitching shall be overstitched not less than 1½ inches at each break. Thread tension shall be maintained so that there will be no loose stitching. The looper thread shall be on the unexposed surface of fabric in the finished bag.

3.2.1.3 *Bartacking.* Bartacking shall be $\frac{5}{8} \pm \frac{1}{16}$ inch in length and shall contain 28 stitches.

3.2.2 *Seam types.* Seams shall be formed as shown on drawings, with seam allowance and seams sewn so that no raw edges, runoffs, twists, pleats, puckers, or open seams shall result.

3.2.3 In all sewing operations, the smallest practicable needle size shall be used. Channels shall not be obstructed at any seamed point such as to prevent a free flow of feathers and down throughout all channels.

3.2.4 *Piecing.* In order to utilize short pieces, the webbing guards, binding strips, weather strips and materials which are unexposed in the finished bag may be pieced once only in either length or width. Piecing of the 2½-inch tape or the 1-inch webbing shall be overlapped not less than ¾ inch and boxstitched so that it will not interfere with the operation of the slide fastener. Piecing of the 1½-inch webbing or the ¾-inch tape will not be permitted.

3.2.4.1 *Piecing of fur strips.* In order to economically utilize class 2 fur strips, piecing will be permitted as required. Seaming will be accomplished in accordance with seaming requirements specified in Specification MIL–F–11861.

3.2.5 *Filling bag and weather strip.* The weight of each unfilled bag (including insulating pads, drag pad and fur ruff) shall be recorded in ink or indelible pencil on an attached commercial type tag. The sides and bottom section, weather strip, and top section shall be filled with the materials and in the quantities shown on drawings. During the filling operation, each bag shall be weighed three times as follows: After filling:

(a) Sides and bottom section

(b) Weatherstrip

(c) Top section

The weight of the filling material added by each operation shall be recorded on the tag. The contractor shall not remove the weight tag.

3.2.6 *Beating and cleaning.* The completed bag shall be well beaten out so that the filling material is evenly distributed throughout all channels. All loose ends of thread shall be trimmed and any spots or stains removed. The finished bag shall be cleaned of all feathers and down.

3.2.7 *Assembling.* Each finished bag shall be fitted with a completed and closed liner, secured in the bag by the tie tapes and loops provided. Tapes shall be so tied as to make the liner readily removable. After the liner has been properly attached, the slide fastener shall be completely closed.

3.2.8 *Workmanship.* The finished evacuation bag shall conform to the quality and grade of product established by this specification. Occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Inspection for acceptance. Inspection shall be in accordance with Standard MIL–STD–105, unless otherwise indicated.

4.2.1 *Inspection of components.* The quality assurance provisions for components, except as otherwise required by this specification, shall be in accordance with the applicable documents referenced herein to the extent applicable. Where there is a conflict in quality assurance provisions, this specification shall govern. In addition, inspection for characteristics shown shall be made of materials listed in table I. The inspection level shall be L–1 and the acceptable quality level (applicable to the individual unit) shall be 6.5 test failures per hundred units.

TABLE I

Component and lot expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Individual unit	Requirement applicable to: Lot average	Number determinations per sample unit	Results reported as	Sample unit
Dynel pile fabric; square yards [1].	Courses per inch.	3.1.16	Visual	.....	X	1	Pass or fail.	
	Wales per inch.	3.1.16	Visual	.....	X	1	Pass or fail.	
	Denier of staple.	3.1.16	Standard commercial.	.....	X	.....	Pass or fail.	One piece, 24 by 24 inches.
	Weight per square yard.	3.1.16	Standard commercial.	.....	X	.....	Pass or fail.	
	Fabric thickness.	3.1.16	(See 3.1.16).	X	.....	1	Pass or fail.	
	Dynel content.	3.1.16	(See 4.3.1).	.....	X	5	Pass or fail.	
Slide fastener tape; feet.	Fastness to laundering.	3.1.17.4	Specification CCC–T–191, method 5610 [2].	X	.....	1	Pass or fail.	One foot
Wire; feet [1].	Material identification.	3.1.21	Standard commercial.	.....	X	.....	Pass or fail.	
	Diameter	3.1.21	Standard commercial.	.....	X	.....	To nearest 0.001 inch.	

[1] Will be accepted on the basis of contractor's certification of compliance with requirements specified. The Government reserves the right to check test components to determine validity of contractor's certification.

[2] The test specimen shall be the full width of tape 4 inches in length.

4.2.2 *Process inspection.* Process inspection shall be made at any point or during any phase of the manufacturing processes to determine whether construction details which cannot be examined in the finished product are in accordance with requirements indicated in the specification and drawings. Care shall be taken to see that cut parts conform to contours and dimensions indicated on drawings, and that the dynel pads are assembled to bottom portion of bag as specified in section B–B, Drawing 2–3–126. The Government reserves the right to exclude from consideration any material or service for which process inspection has indicated nonconformance.

4.2.3 *Intermediate inspection.* Inspection of the unfilled bags shall be made to verify the weight of individual casings (without liners) re-

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corded by the contractor on the tag attached to each bag. This inspection shall be performed prior to filling bags and weatherstrips with down, and feather and down mixture (see 3.2.5). The sample unit for this examination shall be one unfilled bag.

4.2.3.1 *Examination for weight of unfilled casing and dimensions of cut parts.*

Examine	Defect	Classification		
		Major	Minor	
			A	B
Weight of unfilled casing	More or less than weight indicated on bag:			
	—by two ounces or less		X	
	—by more than two ounces	X		
	Note. Utmost care shall be taken to see that scales used to perform this examination are properly calibrated and register accurate weight.			
Dimensions of cut parts (bottom, side and top sections, and dynel pad).	*Any deviation exceeding the minus tolerance* specified on drawings, but not exceeding twice the minus tolerance.			X
	Any deviation exceeding twice the minus tolerance specified on drawings.		X	
	Any deviation exceeding three times the minus tolerance specified on drawings.	X		
	Any deviation exceeding the plus tolerance specified on drawings.			X

4.2.3.2 *Filling inspection.* Inspection of partially filled and filled casings (see 3.2.5) shall be made to verify the weight of the filling material blown in. Inspection shall be made of the sample unit after each of the filling operations. The sample unit for these examinations shall be one casing selected immediately after each filling operation.

4.2.3.2.1 *Examination for weight of filling material.*

Examine	Defect	Classification		
		Major	Minor	
			A	B
Weight of filling, in sides and bottom section.	One ounce or less under the minimum specified.		X	
	More than one ounce under minimum specified.	X		
Weight of filling, in top section.	½ ounce or less under the minimum specified		X	
	More than ½ ounce under minimum specified	X		
Weight of filling, in weatherstrip.	Less than minimum specified on drawings.	X		
	Note 1: The weight of filling material shall be as specified on drawings.			
	Note 2: To determine the weight of filling material added by each operation (see 3.2.5) subtract the total weight before from the total weight after each operation.			
	Note 3: Inspection for weight of filling material shall be performed immediately following each operation. No sample shall be selected for inspection for weight if more than one hour has elapsed between the filling and the inspection.			

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4.2.4 *Examination of the end item.* The classification of defects found during examination shall be in accordance with 4.2.4.1., 4.2.4.2. and 4.2.4.3. The sample unit for these examinations shall be one completely fabricated and filled bag with liner.

4.2.4.1 *Examination for visual characteristics of evacuation bag and liner.*

Examine	Defect	Classification		
		Major	Minor	
			A	B
Fabric	Any hole, cut or tear	X		
	Fabric defects:			
	—seriously affecting serviceability or appearance.	X		
	—affecting serviceability or appearance but not seriously.		X	
Webbing	Cut or frayed edges; not firmly or tightly woven	X		
	Any piecing of 1½-inch webbing or ¾-inch tape		X	
Seams and stitching:				
Open seams:				
On channel stitching	Stitch type 301:			
	—more than ½-inch, but not more than 1 inch			X
	—more than 1-inch		X	
	Stitch type 401:			
	—any open seam	X		
On seams that would allow escape of down, or feathers and down mixture from the bag.	Any open seam	X		
On all other seams	½-inch or less			X
	More than ½-inch, but not more than 1-inch		X	
	More than 1-inch, not affecting serviceability		X	
	More than 1-inch, affecting serviceability	X		
	Note. A seam shall be classified as open when one or more stiches joining a seam are broken or when two or more continuous skipped stitches or run-offs occur. On double stitched seams, a seam shall be considered open when either one or both sides of the seam are open.			
Raw edges (when securely caught in stitching).	½-inch or less			X
	Note. Raw edges not securely caught in stitching shall be classified as open seams.			
Run-offs (see open seams)				
Seam and stitch type	Wrong seam or stitch type	X		
Stitch tension	Loose tension, resulting in a loose seam		X	
	Loose tension on channel stitching resulting in loosely exposed top or bobbin thread.			X
	Tight tension (stitches break when normal strain is applied to seam or stitching).		X	
	Note. Puckering is evidence of tight tension. When puckering is evident, test seam by exerting normal pull on puckered areas.			
Stitches per inch	One stitch less than minimum specified			X
	Two or more stitches less than minimum specified.		X	
	One or more stitches in excess of maximum specified.			X
	Note. Variation in the number of stiches per inch is to be scored only when variation on a major portion of the seam exists.			

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Examine	Defect	Classification		
		Major	Minor	
			A	B
Stitching margin; stitching gage.	Not as specified, affecting serviceability		X	
Stitching ends	Backstitched less than ½-inch (except where turned under in a hem or held down by other stitching).		X	
Thread breaks	On stitch type 301:			
	—Backstitched less than 1-inch		X	
	On stitch type 401:			
	—Overstiched less than 1½-inches		X	
	Note. Thread breaks not backstitched or not overstitched shall be classified as open seams.			
Components and assembly	Seams badly puckered, twisted or pleated			X
	Needle chews	X		
	Note. Needle chews (caused by run-offs or repairs) which do not weaken the fabric shall not be considered defects.			
	Any component or required operation omitted (unless otherwise classified herein).	X		
Piecing:				
Casing and weatherstrip	Any component of casing or weatherstrip not pieced as specified or pieced more than once.		X	
Fur strip	*Method of piecing not as specified in Specification* MIL–F–11861.		X	
Channels (see drawings for spacing of channels).	Number of channels less than specified	X		
	Any channel trapped, causing part of channel to remain unfilled.	X		
	Channels vary from specified width by more than ½-inch.		X	
Filling	Any unfilled channel or weatherstrip	X		
	Any partially filled channel or weatherstrip		X	
	Unevenly filled channels or weatherstrip			X
Content label and marking	Any required marking or content label omitted, *illegible or incorrect.*		X	
	Misplaced			X
Letters U. S.	Omitted, illegible, misplaced or not specified type or size.			X
Cleanness	Spots or stains. Thread ends or clinging feathers and down not removed throughout bag.			X
Details of contruction				
Face opening	Wire stiffener not securely attached to wire retainer.		X	
	Fur piece not securely caught in face opening seam.		X	
Slide fastener	Slider jams or *fails to interlock chain scoops*	X		
	Any portion of fastener broken, bent, missing or not aligned, making fastener unusable.	X		
Assembly (Slide fastener).	Thong omitted		X	
	Finished length of thong less than 3-inches or more than 4½-inches.			X
	Slide fastener tape not assembled to bag as shown on drawings:			
	—seriously affecting serviceability	X		
	—affecting serviceability, but not seriously		X	
	Stitched too close to metal chain, preventing slider *from sliding freely.*	X		

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Examine	Defect	Classification		
		Major	Minor	
			A	B
	Slide fastener tapes stitched to webbing strips too loose, causing a noticeable bulge which prevents slide fastener from closing easily.		X	
	Edges of casing not securely caught in both rows of stitching through fastener assembly for a distance of more than 3-inches.		X	
Bartacks	One or more omitted or loose	X		
	Not sewn through materials as shown on drawings.		X	
Weatherstrip	Not assembled to top section as shown on drawings.	X		
	Tube portion of weatherstrip caught in slide fastener assembly stitchings or in bartacks.	X		
Loops	Not formed as shown on drawings		X	
	One or more omitted; not securely caught in seam.		X	
	Any loop set more than 1-inch from dimensions specified on drawing.			X
Hardware	Broken or malformed; finish omitted; corroded areas; burrs or sharp edges which may damage the item or cause injury in handling.	X		
Liner				
Tie tapes.	Not securely bartacked to liner	X		
	Side tapes not bartacked through reinforcement as shown on drawing.	X		
	Mislocated by ½-inch or more, but not more than 1-inch.			X
	Mislocated by more than 1-inch		X	
	Assembled with coated side of fabric on outside	X		
Snap fasteners	Clinched excessively tight, cutting surrounding fabric.	X		
	Loosely clinched, permitting cap or stud to rotate.			X
	Male or female fastener misplaced, failing to effect a smooth and secure closure.	X		
	Any malfunction of snap fasteners	X		
Carrying straps (A and B)	Any strap not one piece construction		X	
	Handles not securely boxstitched as specified:			
	—seriously affecting serviceability	X		
	—affecting serviceability, but not seriously		X	
Drag pad	Misplaced:			
	—seriously affecting serviceability	X		
	—affecting serviceability, but not seriously		X	

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4.2.4.2 *Examination for dimensions of liner.*

Examine	Defect	Classification	
		Major	Minor
Dimensions	Overall length:		
	—less than 7 feet, 9 inches		X
	—less than 7 feet, 8 inches	X	
	Width at any location less than twice the minus tolerance indicated on drawing.	X	

4.2.4.3 *Examination of carrying bags.* Defects found during examination shall be classified in accordance with classification of defects contained in Specification MIL–B–3108. In addition thereto, the carrying bag shall be examined for the following characteristics:

Examine	Defect	Classification	
		Major	Minor
Handles	Either handle omitted	X	
	Not boxstitched as specified		X
	Misplaced:		
	—by more than 1 inch	X	
	—½ inch or more, but not more than 1 inch		X
Reinforcements	Any reinforcement omitted	X	
Flap	Omitted	X	
	Set under or over specified location by more than ½ inch		X
Grommets	Size or type not as specified:		
	—seriously affecting serviceability	X	
	—affecting serviceability, but not seriously		X
	Improperly set, clinched loosely, allowing grommet to rotate in hole.		X
Drawcord	Any cuts or abrasions	X	
	Ends not seared		X
	Less than required length:		
	—by one inch or more, but less than 2 inches		X
	—by 2 inches or more	X	

4.2.4.4 *Inspection levels and acceptable quality levels.* The inspection levels and acceptable quality levels (AQI's) (expressed in defects per hundred units) shall be as follows:

Examination	*Inspection level*	*AQL's*
4.2.3.1—Examination for weight and dimensions of cut parts.	L–4	1.0 Major 4.0 Total Major and Minor A 10.0 Total Major, Minor A and Minor B defects combined.
4.2.3.2.1—Examination for weight of filling material.	L–4	2.5 Major 3.5 Total Major and Minor A
4.2.4.1—Examination for visual characteristics of evacuation bag and liner.	II	2.5 Major 10.0 Total Major and Minor A 65.0 Total Major, Minor A and Minor B defects combined.
4.2.4.2—Examination for dimensions of liner.	L–6	4.0 Major 10.0 Major and Minor defects combined.
4.2.4.3—Examination of carrying bag.	As specified in MIL–B–3108.	As specified in MIL–B–3108

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4.2.5 *Examination of preparation for delivery requirements.* An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packed, selected just prior to the closing operation. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be L–4 and the AQL shall be 4.0 defects per hundred units.

Examine	Defect
Markings (exterior and interior.)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing.
	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of containers.
Content	Number per container is more or less than required.

4.3 Tests. The methods of testing specified in Specification CCC–T–191 shall be followed wherever applicable.

4.3.1 *Dynel content.* A specimen approximately 1 gram in weight is dried at 221° to 230°F. (105° to 110°C.) to a constant weight. Place the specimen in a Soxhlet or similar extractor and extract with acetone until the solution runs clear (a yellow coloration indicates the presence of dynel). Dry the specimen to constant weight as above and calculate the percentage of dynel as follows:

$$\frac{\text{Weight of original dry specimen—weight of dry reside} \times 100}{\text{Weight of original dry specimen}}$$

Final result shall be the average of five specimens tested as above.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* Each evacuation bag, assembled as specified, shall be laid out flat and then tightly rolled down from the hood end. The completed roll shall be secured with cotton tape. Each rolled evacuation bag shall be placed within the carrying bag component and the bag closure effected by means of the tie cord.

5.1.1.2 *Intermediate packaging.* Each evacuation bag, until packaged as specified in 5.1.1.1, shall be placed in a snug-fitting polyethylene bag. The polyethylene flexible film shall be 0.003 inch thick conforming to type I, grade A, B, or C, finish 1 of Specification L–P–00378. All seams and closures shall be effected by heat sealing and shall conform to heat seal strength requirement in Specification L–P–00378. Prior to or during the final heat sealing closure operation, excess air within the bag shall be expelled. Alternatively, the bag mouth may be closed by means of a mechanical tie (plastic or paper coated annealed wire).

5.1.2 *Level C.* Evacuation bags shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 *Level A.* One complete evacuation bag, packaged as specified in 5.1, shall be packed within a fiberboard shipping container conforming to class 2, V3c, style OSC–L of Specification PPP–B–636, except that the inner flaps meet at the center, all flaps the same length, and the outer flaps overlap as may. The inside of the container shall be provided with a one-piece double-faced corrugated liner having a minimum bursting strength of 275 psi conforming to Specification PPP–B–636, abutting at the opposite corner from the manufacturer's joint. Inside dimensions of the shipping container (outside of liner) shall be 23½ inches in length, 15 inches in width and 15 inches in depth. Each container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. In addition, all

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seams, corners, and manufacturer's joint of the shipping container shall be sealed by the application of 3-inch minimum width cloth tape conforming to type II, class 1, olive drab color of Specification PPP–T–60.

5.2.1.1 *Unitizer pack.* When specified, three level B shipping containers of evacuation bags shall be overpacked on end three in length within a fiberboard shipping container conforming to class 2, V3c or V3s, style FOL of Specification PPP–B–636. Inside dimensions of the unitizer shipping container shall be 46¾ inches in length, 16¼ inches in width, and 24 inches in depth. The requirements for closure and strapping shall be in accordance with the appendix of the container specification, except that the flaps shall be glued and only flat steel straps used. In addition, all seams, corners, and manufacturer's joint of the container shall be sealed with 3-inch minimum width olive drab color cloth tape conforming to type II, class 1 of Specification PPP–T–60.

5.2.2 *Level B.* One complete evacuation bag, packaged as specified in 5.1, shall be packed within a fiberboard shipping container conforming to type I (B-flute), class 1, style OSC–L of Specification PPP–B–636, except that the inner flaps meet at the center, all flaps the same length, the outer flaps overlap as may and the manufacturer's joint shall be stitched. The minimum bursting strength of the fiberboard shall be 350 psi. The inside of each container shall be provided with a one-piece double-faced corrugated liner having a minimum bursting strength of 200 psi conforming to Specification PPP–B–636, abutting at the opposite corner from the manufacturer's joint. Inside dimensions of the shipping container (outside of liner) shall be 23½ inches in length, 15 inches in width and 15 inches in depth. All flaps of the fiberboard shipping containers shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive.

5.2.3 *Level C.* Evacuation bags, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be packed in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Standard MIL–STD–129. When level B fiberboard shipping containers are overpacked, thereby becoming inner containers, the marking shall be in accordance with the requirements for interior packages.

5.3.1 *Polyethylene bagged packages.* Polyethylene bagged packages shall have the required information legibly printed or stamped in black on a white paper label placed inside the bag so as to provide ready identification. The white paper label shall contain the following information:

(a) Stock number
(b) Nomenclature
(c) Quantity

6. NOTES

6.1 Intended use. The bag, with liner, covered by this specification is for military personnel of the Department of Defense for the evacuation of sick or wounded, or otherwise to provide protection for an individual casualty under conditions of extreme cold.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2) and whether the unitizer pack shall be specified (see 5.2.1.1).

6.3 Standard and shade samples. For access to standard samples of thread and shade samples of various materials, address the procuring office issuing the invitation for bids.

6.4 The sample of nylon duck furnished as standard for shade, finish and colorfastness was dyed with neutral pre-metallized dyes using the following formulation. This information is not a

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mandatory requirement and is supplied for information only.

Premetallized olive green 2BL
Premetallized yellow 2RL
Premetallized brown 3BL
Acid black 58

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Custodians:
Army—Quartermaster Corps
Navy—Bureau of Medicine and Surgery
Air Force—Air Force

Other interest:
Army—M

Preparing activity:
Army—Quartermaster Corps

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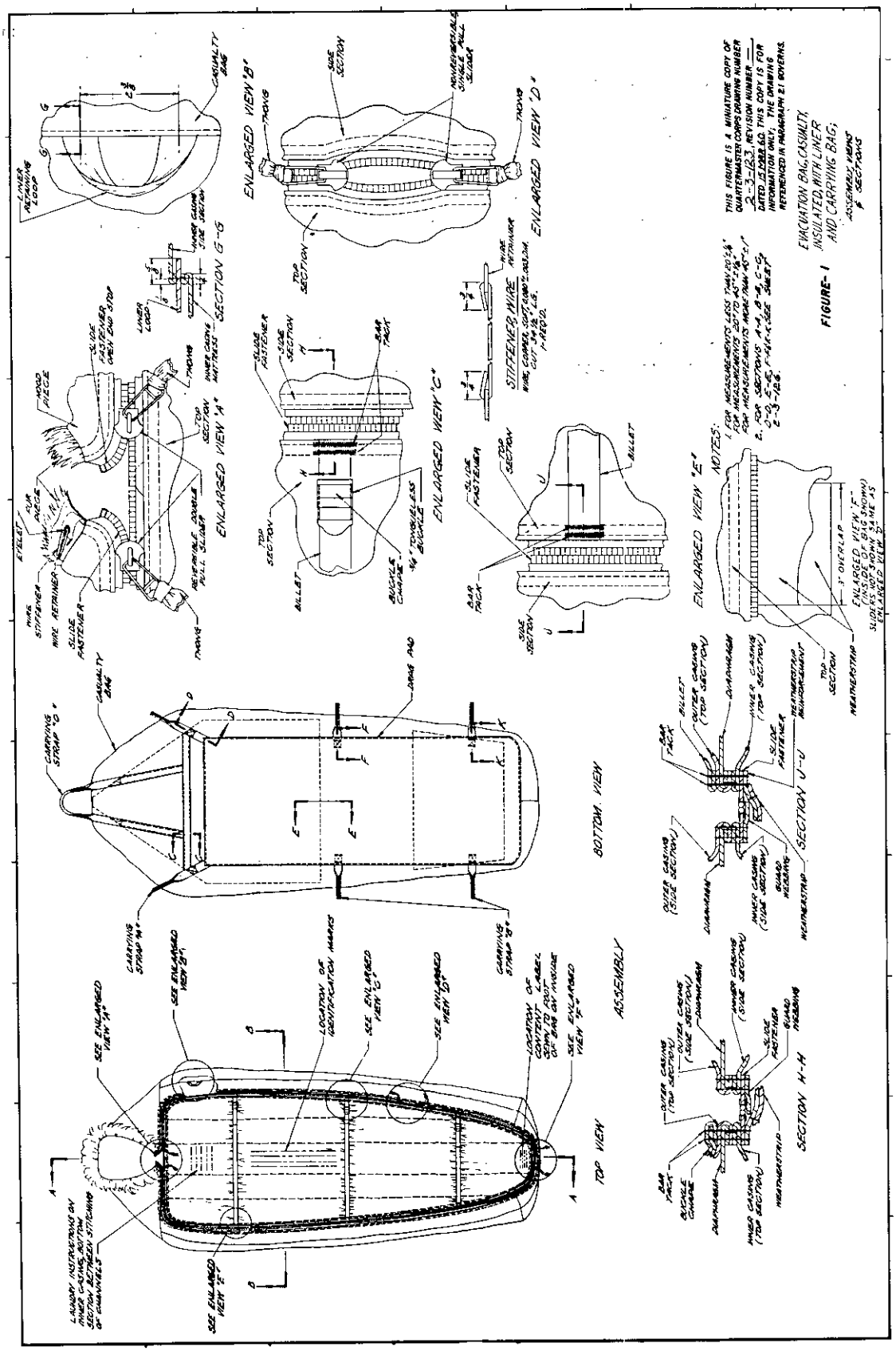


FIGURE 1. *Evacuation bag, casualty, insulated, with liner and carrying bag; assembly, views and sections.*

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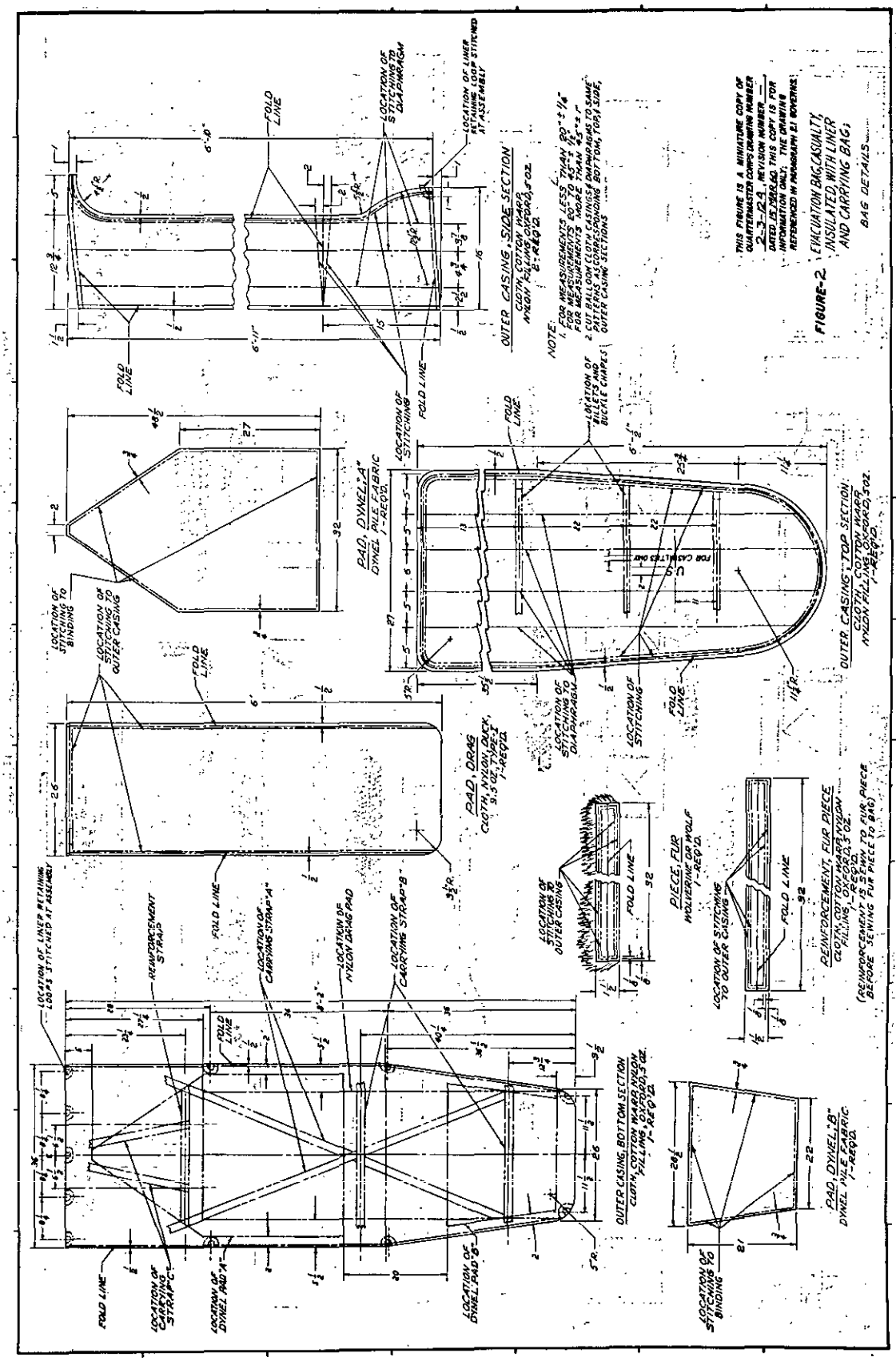


FIGURE 2. *Evacuation bag, casualty, insulated, with liner and carrying bag; bag details.*

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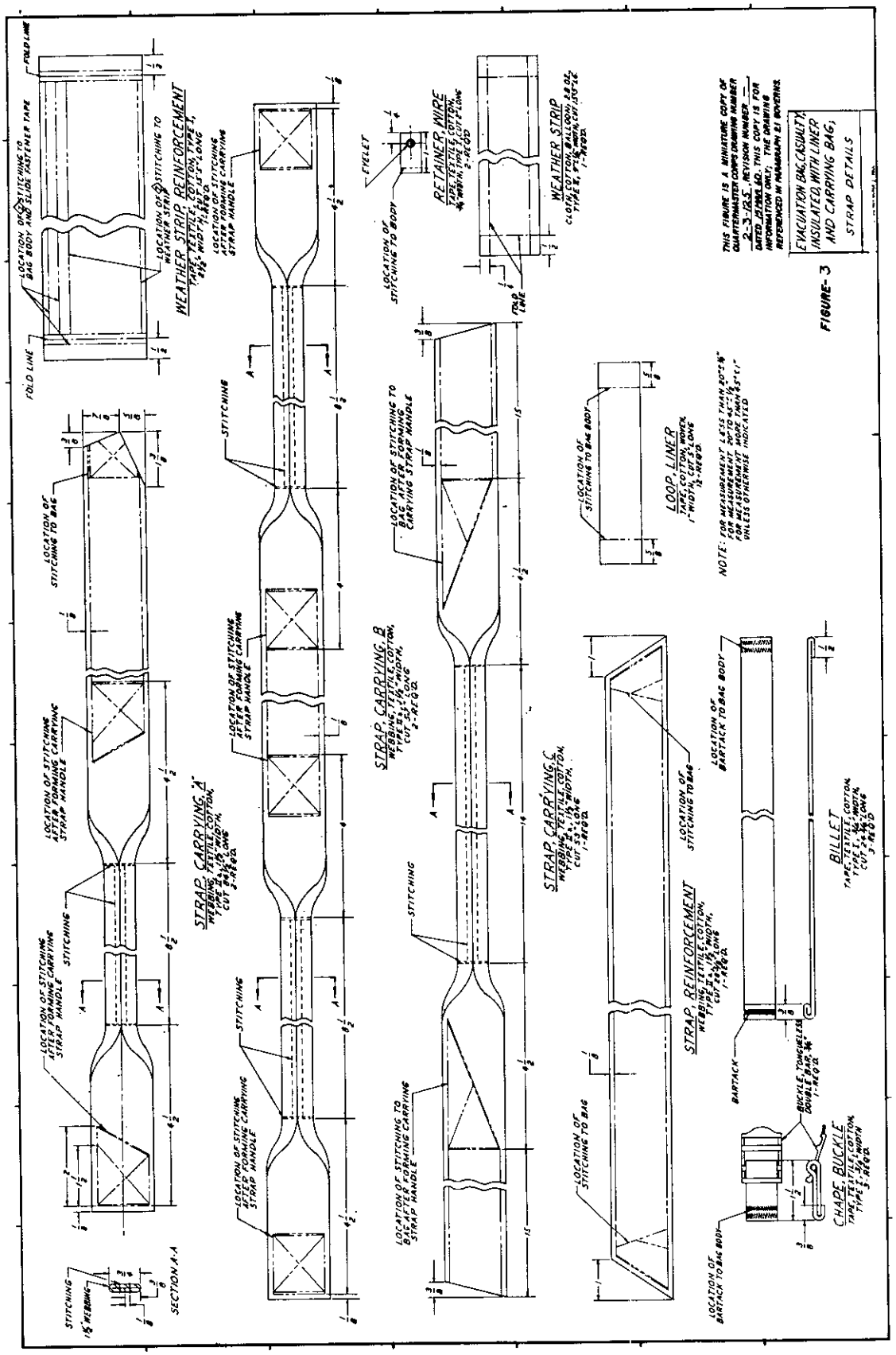


FIGURE 3. *Evacuation bag, casualty, insulated, with liner and carrying bag; strap details.*

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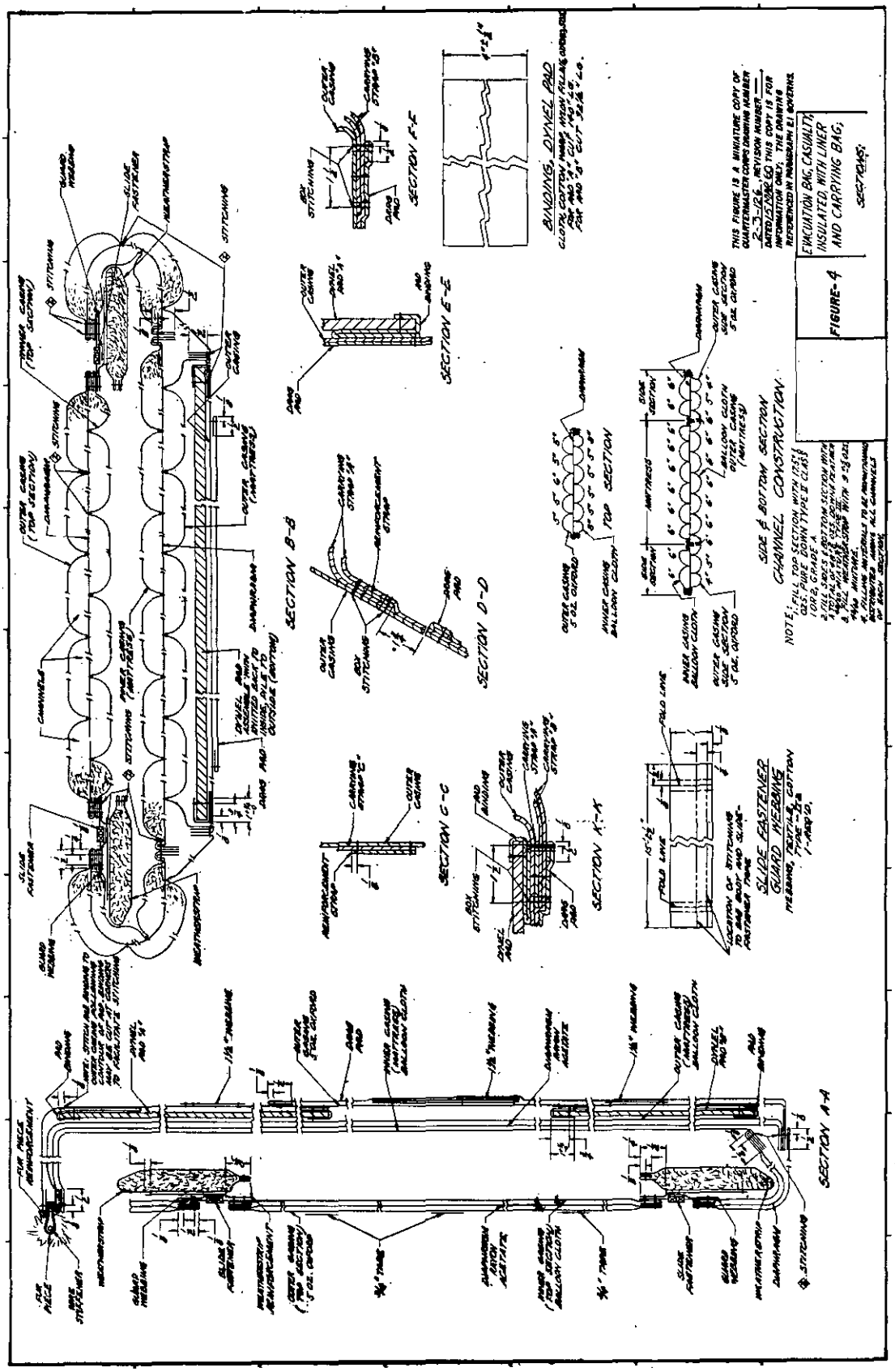


FIGURE 4. *Evacuation bag, casualty, insulated, with liner and carrying bag; sectors.*

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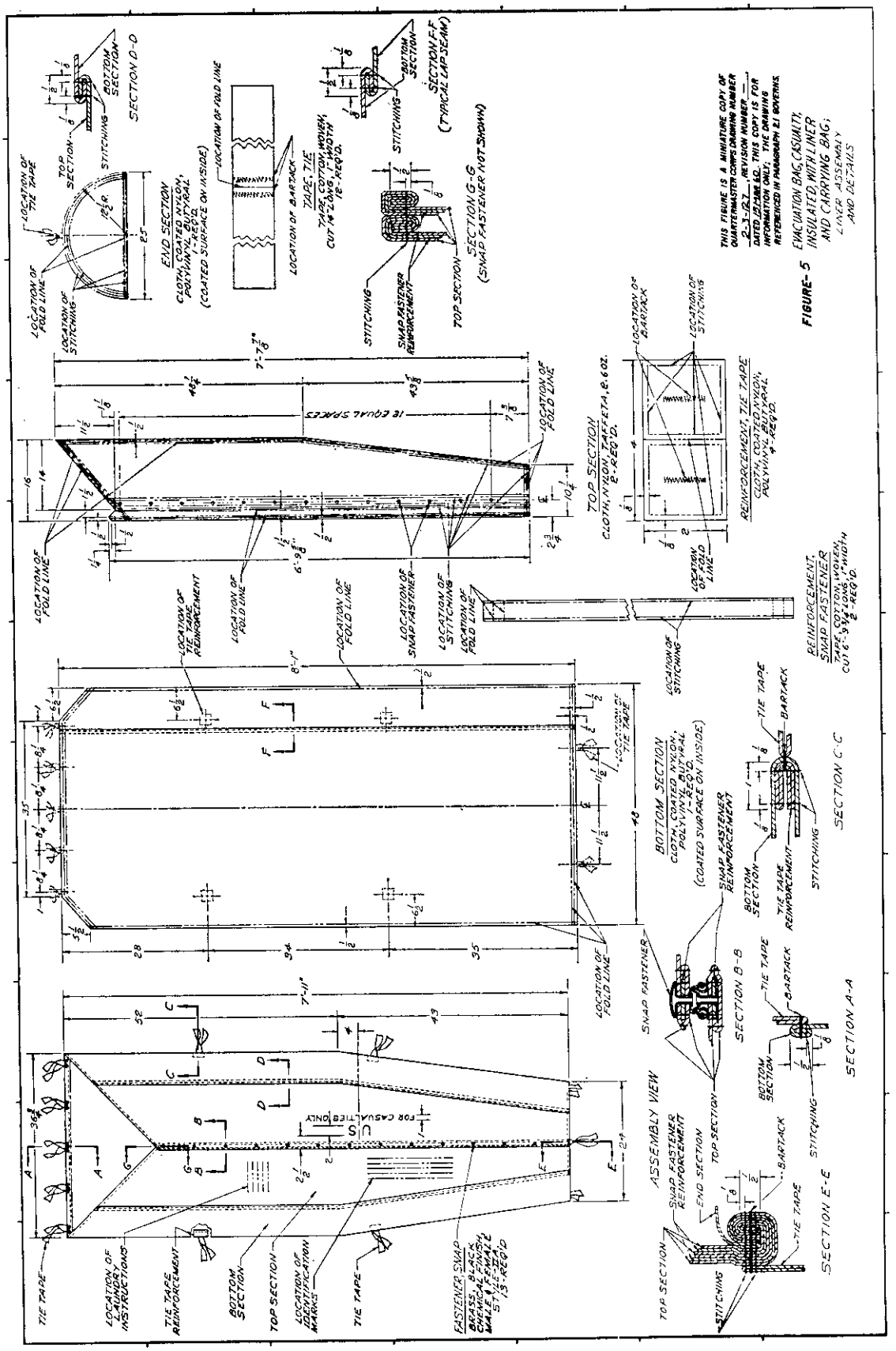


FIGURE 5. *Evacuation bag, casualty, insulated, with liner and carrying bag; liner assembly and details.*

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